

DEVIKA NAIR

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M.S. ECE student at the University of Washington with 3+ years of embedded firmware experience at SanDisk (NAND/SSD, eMMC/UFS). Strong in low-level C/C++, RTOS, FPGA/RTL, and hardware bring-up and debug. Seeking a Fall 2026 embedded / hardware internship.

EDUCATION

University of Washington

Seattle, WA

M.S., Electrical and Computer Engineering

Sept. 2025 – Jun. 2027 (Expected)

- **Relevant coursework:** Digital System Design–FPGA; Computer System Architecture; Embedded & Real-Time Systems; Deep Learning for Embedded/Real-Time Intelligence; VLSI Design.

MKSSS Cummins College of Engineering for Women

Pune, India

B.Tech., Electronics and Telecommunications

Jul. 2017 – Jun. 2021

- **Relevant coursework:** VLSI Design; Microcontrollers & Embedded Systems; Digital Signal Processing; Computer Networks; Data Structures & Algorithms; Control Systems.

EXPERIENCE

Graduate Teaching Assistant — EEP 524, Applied High-Performance GPU Computing

Seattle, WA

University of Washington, ECE (Prof. Colin Reinhardt)

Sep 2026 – Dec 2026

- Develop course materials, grade assignments, and provide student support to enhance learning outcomes in a graduate GPU-track course.
- Support instruction in CUDA/OpenCL kernel programming, GPU/many-core microarchitecture, memory subsystems (cache, multi-banked, interconnects), ISA/assembly, and parallel C/C++ development.

SSD Firmware Engineer Intern

Milpitas, CA

SanDisk

Jun 2026 – Sep 2026

- Developing firmware telemetry and physical storage algorithms for enterprise SSD products.
- Applying prior NAND flash, reliability, and firmware experience (eMMC/UFS, CVD analytics) to enterprise SSD (NVMe/PCIe) development.

Research Assistant — UW SEAL Lab (Prof. Alex Mamishev)

Seattle, WA

University of Washington

Winter 2026 – Spring 2026

- Built a real-time embedded sensing system for gas-leak detection on ESP32-S3 (Xtensa LX7) in C/RTOS: interrupt-driven, DMA-based acquisition over SPI/I²C for kHz-rate MEMS data, with a low-latency DSP pipeline (FFT, RMS, feature extraction) and end-to-end firmware through WiFi/LoRa transmission.

Firmware Engineer → Senior Firmware Engineer

Bengaluru, India

SanDisk (Western Digital)

Jul 2021 – Jul 2025

- Drove end-to-end NAND flash (eMMC/UFS) feature implementations for a major automotive client, enabling on-time launch (\$45M revenue, zero customer-reported issues); improved NAND endurance by 25% and reduced latency variation by 15%.
- Built a Cell Voltage Distribution (CVD) analytics framework integrating read-retry and ECC telemetry, enabling voltage-drift modeling and accelerating root-cause debug by ≈40%.
- Contributed to ASPICE CL3 certification (an industry first in embedded storage) and published a [U.S. Patent](#) on predictable low-latency NAND access; delivered a firmware MVP unlocking \$10M in revenue.
- Supported bring-up, validation, and MISRA-C compliance for automotive-grade NAND reliability features.

TECHNICAL SKILLS

Languages: C, C++, Python, Verilog, Bash/Shell, MATLAB, SQL

Embedded & Firmware: ARM Cortex-M/A; bare-metal/low-level C; RTOS (FreeRTOS, ThreadX); interrupts, DMA; Linux drivers, U-Boot, linker scripts; NAND/SSD eMMC/UFS; reliability (Refresh Manager, EPWR, CVD); MISRA-C/Parasoft; bring-up, validation, unit testing

Hardware / FPGA / EDA: Verilog RTL, Vivado, ModelSim; VLSI fundamentals; JTAG/SWD; logic analyzers, oscilloscopes

Build & Tools: Git, GCC, GDB, Make, CMake, Jenkins, Jira, VS Code, Vim

Protocols: I²C, SPI, UART, eMMC, UFS, PCIe, NVMe, TCP/IP

PROJECTS

Context-Switch Latency Characterization on Raspberry Pi 4 | C, Linux, POSIX, ARM

2025

- Benchmarked Linux scheduling on ARM Cortex-A72 (SCHED_OTHER vs. SCHED_FIFO) with POSIX semaphores in C11, measuring context-switch latency (5.61–10.55 μ s) to bound soft real-time behavior.

Heuristic Write-Amplification Analysis for Flash Storage | MATLAB, Python, NAND FTL Simulation

2025

- Heuristic firmware model to reduce NAND write amplification via adaptive data placement and wear-leveling (no ML dependencies).

LEADERSHIP

Corporate Relations Representative — IEEE Women in Engineering (WIE)

Seattle, WA

University of Washington — lead industry/sponsor outreach for the student board

2025 – 2026